AAGAM Web (Next.js) UI Verification Guide

How to run the Next.js web app and verify its role surfaces through the browser — with a worked, UI-driven reproduction of BUG-028 (merged milk-grid partial due).

Repository	Saikumar-bali/AAGAM_E-commerce
App	`apps/admin-dashboard` (Next.js 15 App Router) — serves Shop, Store, Rider and Admin surfaces
Date	2026-10-11
Database	Sandbox Postgres, `DATABASE_URL=$AAGAAM_DATABASE`
Method	Headless Chromium (Playwright) driving the real UI — no direct API calls

1. What runs where

The web app is one Next.js build (`apps/admin-dashboard`) that serves every browser role by route group. The API gateway is a separate NestJS service; the dashboard proxies `/api/*` to it, so the browser stays same-origin.

Surface	Route group	Example route
Shop (customer)	`(shop)/shop`	`/shop/subscriptions`
Store portal	`(store)/store`	`/store/subscriptions`
Rider portal	`(rider)/rider`	`/rider/runs`
Admin dashboard	`(admin)/admin`	`/admin/riders`
Auth	`(auth)`	`/login`

2. Running it locally

Build the shared packages first, then start the gateway and the dashboard:

```
# 1. shared packages (types, utils, database) — the dashboard imports them
npm run build --workspace=packages/types
npm run build --workspace=packages/utils
npm run build --workspace=packages/database   # also runs prisma generate

# 2. api-gateway on :3005 against the sandbox database
cd apps/api-gateway
set -a && source .env && set +a
export DATABASE_URL="$AAGAAM_DATABASE"
node dist/src/main.js            # after `npm run build --workspace=apps/api-gateway`

# 3. Next.js dashboard on :12000
cd apps/admin-dashboard
npm run build && npx next start -p 12000
```

> **Proxy:** `next.config.js` rewrites `/api/:path*` → `API_BACKEND_URL` (default `http://127.0.0.1:3005`). Keep the gateway on :3005 or set `API_BACKEND_URL` to match.

3. Verifying through the UI

Every check below was performed by loading the real pages in a browser and reading the rendered DOM — never by calling the API directly. The work-host URLs used were:

Link	Purpose
`https://work-1-lkhufuuxeteqzqdl.prod-runtime.all-hands.dev/` (port 12000)	Next.js web app — Shop, Store, Rider, Admin
`https://work-2-lkhufuuxeteqzqdl.prod-runtime.all-hands.dev/` (port 12001)	Mobile react-native-web previews (partners + customer) and the same-origin API proxy

3.1 Login

Open `/login`. The form takes a phone or email and a password (password mode is the default; phone OTP is disabled in this build). A store owner lands on `/store`.

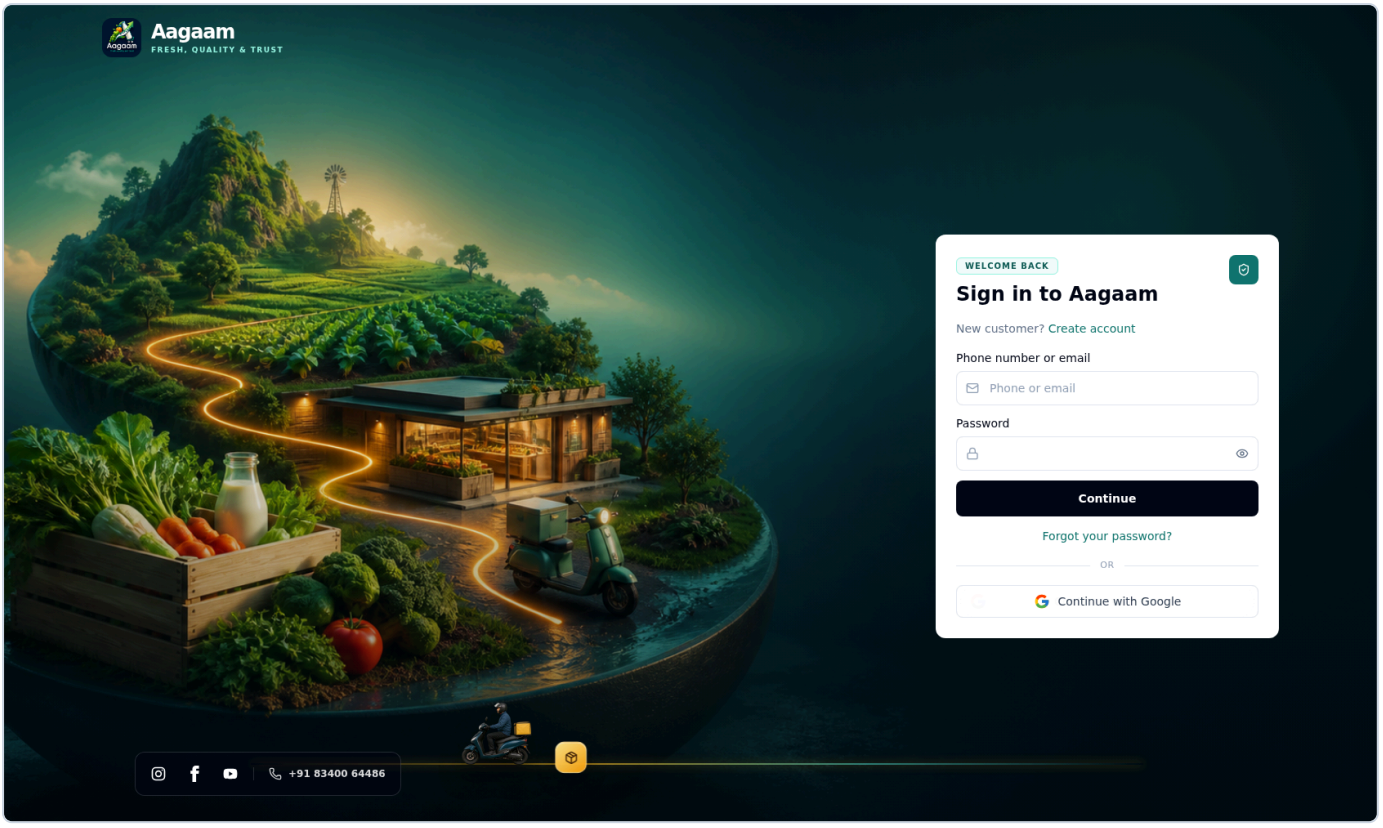


Figure 1 — `/login`: single “Phone or email” field plus password, then *Continue*.

3.2 Store dashboard

After signing in the store workspace loads with live counters (Subscribers, Active plans, Routes today, Cash to count).

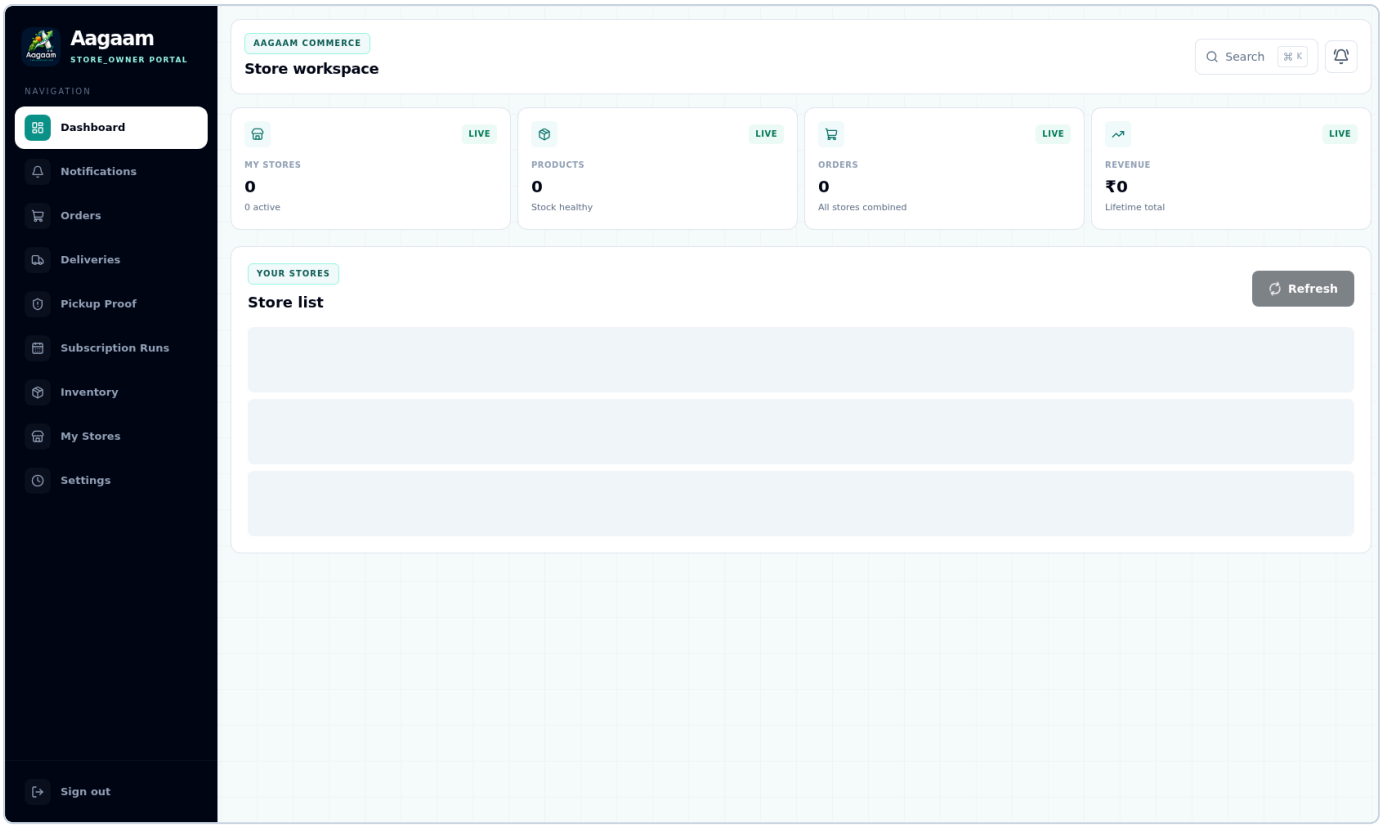


Figure 2 — `/store`: store workspace summary.

3.3 Subscriptions → Milk Grid (Sheet View)

`/store/subscriptions` opens on the **Milk Grid** tab: one row per customer, one cell per day of the month, with **Total L**, **Paid**, **Due** and a **Bill** action. The search box filters by customer, phone or locality.

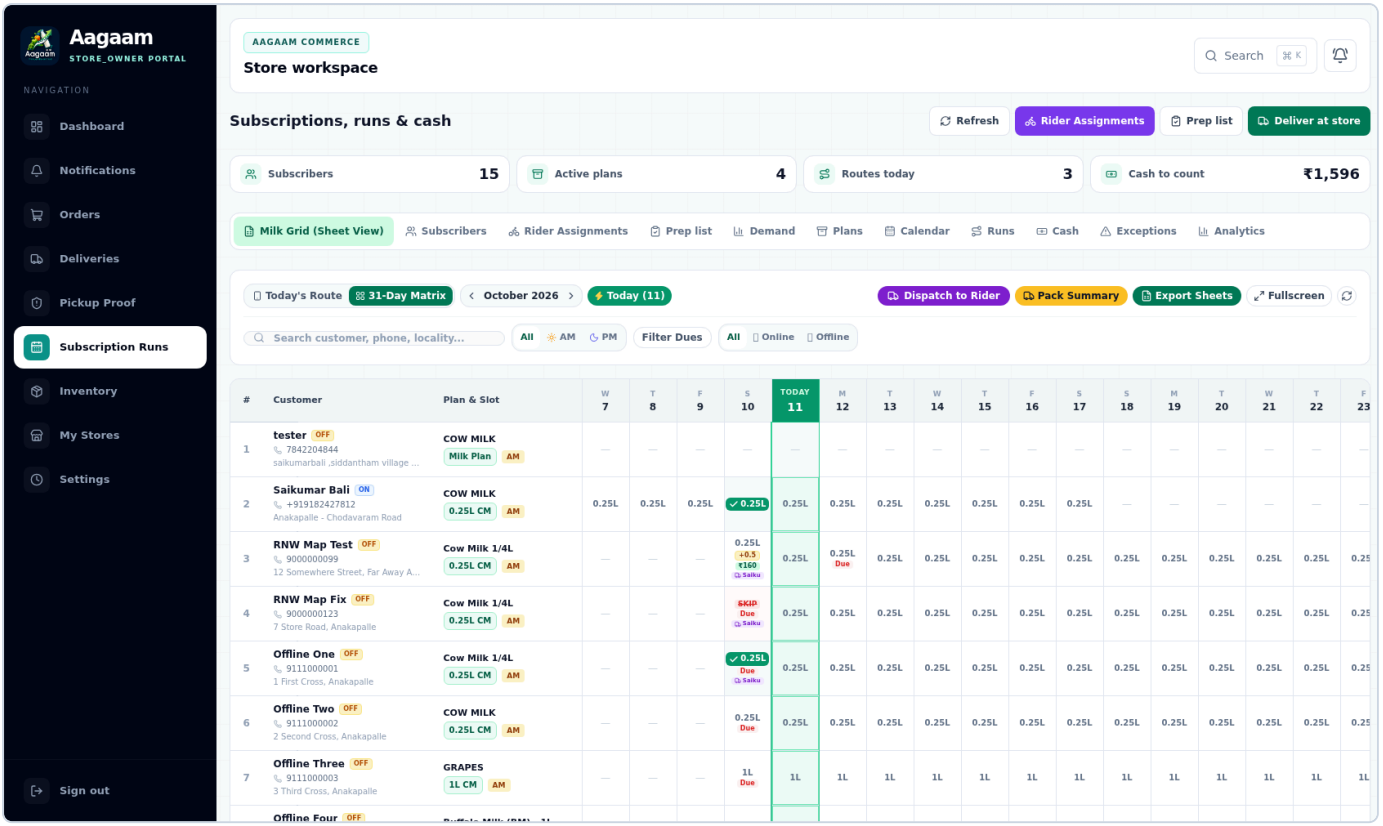


Figure 3 — Milk Grid (Sheet View) with day cells and the Paid / Due / Bill columns on the right.

3.4 Opening the customer statement

Clicking **Bill** on a row opens the *Monthly Bill Statement* modal — the same reconciled money the row shows, plus the WhatsApp message text.

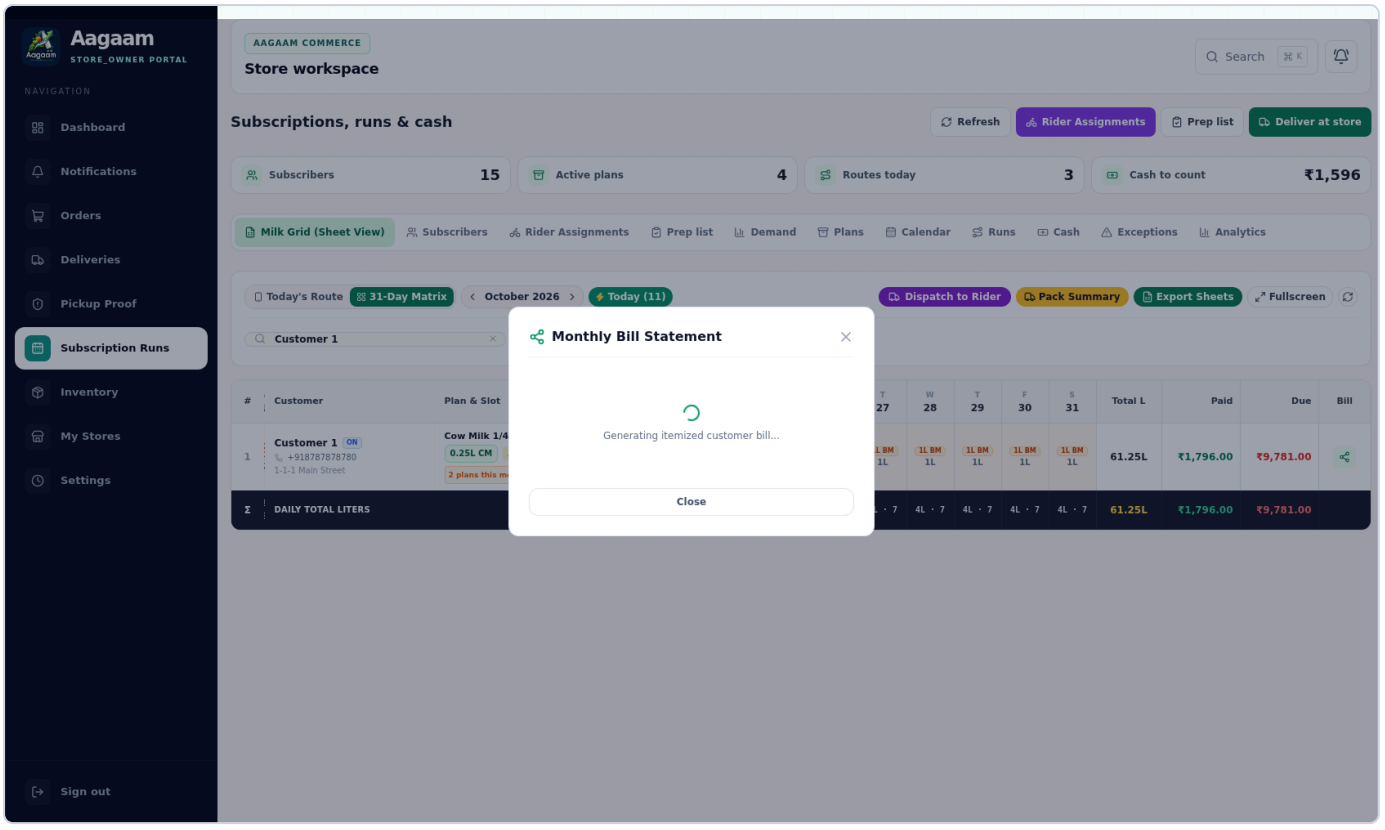


Figure 4 — Monthly Bill Statement modal for the selected customer.

4. Worked example — BUG-028, reproduced through the UI

A customer can hold more than one subscription (a different plan, or the same plan in another slot). The grid **merges** those contracts into a single row, summing litres and money. The bug: when one contract had a delivered day cell that collected only *part* of its due, the merged row reported **Due ₹0.00**.

4.1 Repro steps (UI only)

1. Give a customer two overlapping subscriptions that merge into one grid row.
2. On one of them, leave a delivered day cell under-collected — e.g. a `₹399` day showing `₹200`.
3. Open Store → Subscriptions → Milk Grid and search the customer.

Before the fix. The day-10 cell renders `₹200`, but the row reads **Paid ₹1,796.00 / Due ₹0.00**. The ₹199 the customer still owes for that day is invisible, so the store stops chasing the balance.

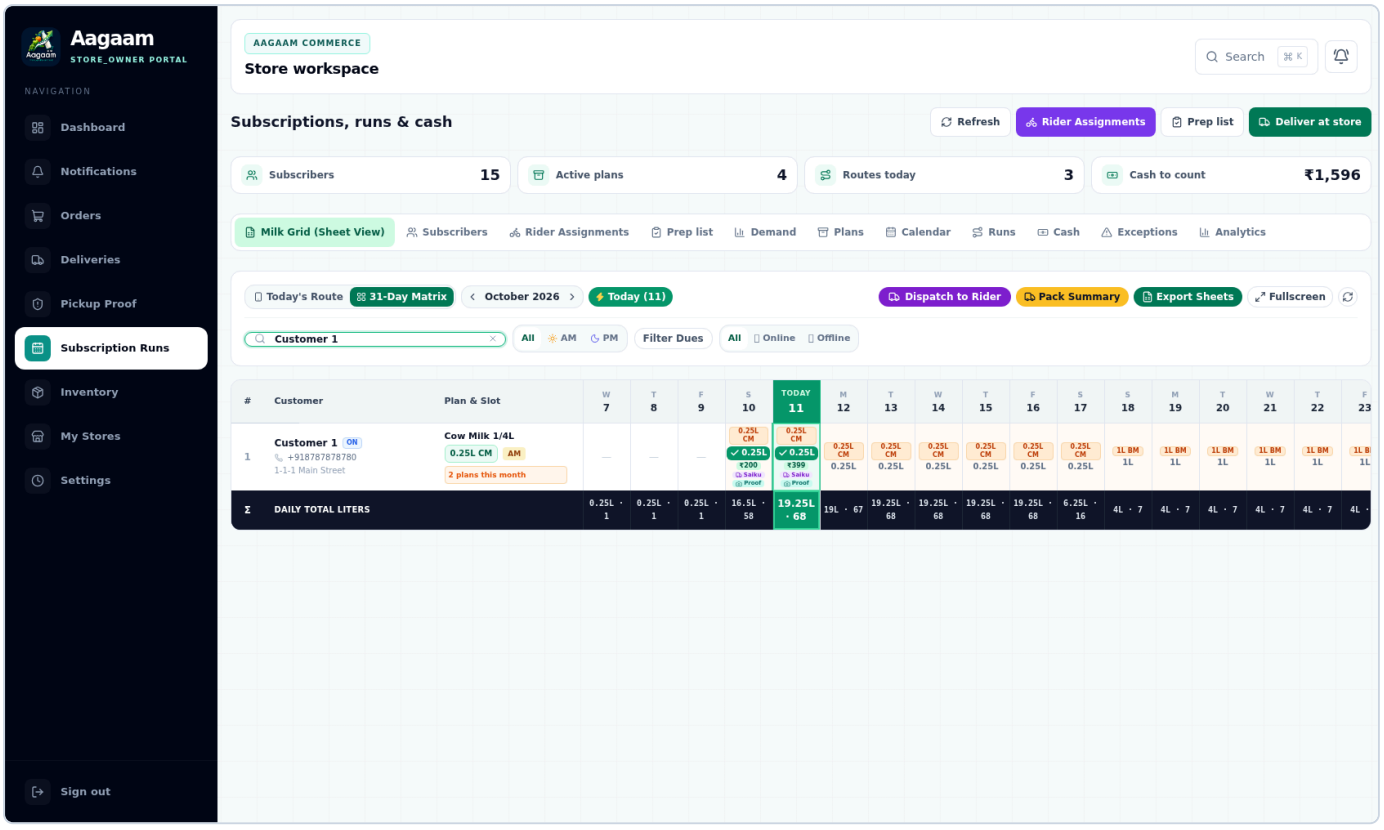


Figure 5 — The day-10 cell shows ₹200 while the row's Due column shows ₹0.00 (pre-fix).

After the fix. The same row now reads **Paid ₹1,796.00 / Due ₹9,781.00** — the under-collected day cell's ₹199 shortfall is surfaced.

4.2 Root cause

`StoreMilkGridService.getGrid()` reconciled the *active* contract's ledger against *every* merged contract's deliveries:

```
reconcileSubscriptionBalance(activeSub, custSubs.flatMap((sub) => sub.deliveries))
```

A sibling contract whose ledger was already fully paid pushed `deliveryCash` above the active ledger, so the helper's `Math.max(0, storedDue - drift)` clamped the under-collected contract's residual to zero.

4.3 Fix

Reconcile each contract against its own deliveries and sum:

```
custSubs.reduce((totals, sub) => {
  const balance = reconcileSubscriptionBalance(sub, sub.deliveries);
  totals.amountCollectedPaise += balance.amountCollectedPaise;
  totals.amountDuePaise      += balance.amountDuePaise;
  return totals;
}, { amountCollectedPaise: 0, amountDuePaise: 0, ... });
```

Each contract keeps its own under-collected cell, while the original ledger-drift absorption still works because it only fires when a ledger is genuinely short of its *own* deliveries.

4.4 Regression test

`store-milk-grid.service.spec.ts` → “merged rows keep each contract's partial due” builds two overlapping contracts (one fully paid, one ₹200-of-₹399) and asserts the row's `totalDuePaise` is **19900**, not 0.

5. Verification checklist

Check	How (UI)	Result
Web app serves	Load `/`, `/login`, `/store/subscriptions`, `/admin`	**200**
Store login	Submit `store@aagam.com` on `/login`	**lands on /store**
Milk grid loads	Read rendered day cells + Paid/Due columns	**live data**
Statement modal	Click **Bill** on a row	**opens**
Console errors	Collect page + console errors during the session	**none** (only the optional Google sign-in 403)
BUG-028	Compare row Due before/after fix	**₹0.00 → ₹9,781.00**

> **Note.** The Google sign-in button logs a harmless `403` / `GSI_LOGGER` console entry when the page origin is not an allowed OAuth origin; it does not affect email/password login.

6. Related material

- **PR #465** — fix(store): keep each merged contract's partial due in the milk grid.
- `skills/aagam-testing/references/bug-register.md` — BUG-028 entry.
- `apps/api-gateway/src/subscriptions/subscription-balances.ts` — the reconciliation helper.